Practice Session 5

19-10-2024

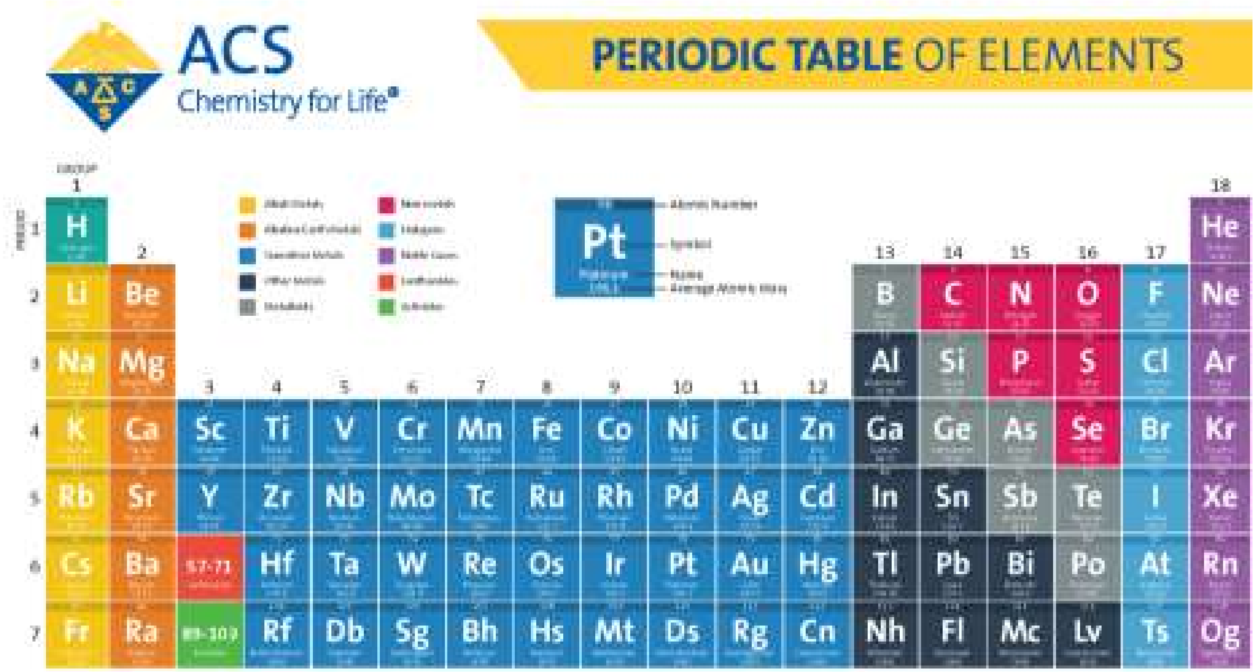
- **1. a.** Predict the trend in first ionization energy between phosphorus and sulfur and account for its trend. 1+2

- The first ionization of Sulfur is lower than that of phosphorus. As we move along a period from left to right Z* increases, size decreases, as a result IE increases. However, here the electronic configuration of phosphorus is [Ne]$3s^2$ $3p^3$ and that of sulfur is $3s^2$ $3p^4$. Due to presence of two electrons in the same 3p orbital of S, they repel each other strongly. This increased repulsion offsets the effect of greater nuclear charge of S compared to P. The half-filled subshell of S+ also contributes to lowering of energy of the ion and hence to the smaller ionization energy.

- **b.** Arrange the following in order of increasing atomic/ionic radius:

2

- i. Cl, Te, Te^{2-}, S
- Cl < S < Te < Te^{2-}
- ii. K^{+}, Ca^{2+}, S^{2-}, P^{3-}
- Ca2+ < K+ < S^{2-} < P^{3-}



- 2. **a.** Rank the electron affinity of the following from smallest to highest: Explain the ranking.
- P, Cl, Ar 2
- Ar < P < Cl
- Ar having the very stable complete shell electronic configuration, do not have any tendency to accept any additional electron.
- From P to Cl, Z* increases. At the same time, Cl ([Ne] $3s^2$ $3p^5$) has one electron short that the stable noble gas electronic configuration. Hence Cl has a very high electron affinity. On the other hand, P having stable half-filled p subshell, ([Ne] $3s^2$ $3p^3$). The additional electron enters one of the already occupied p orbital and experience repulsion. Hence, IE of P < Cl.

- b. Using Slater's rule show that size of Cl^- is larger than Cl. 3
- electronic configuration of Cl is $(1s^2)(2s^2\ 2p^6)(3s^2\ 3p^5)$ and

  of Cl^- is $(1s^2)(2s^2\ 2p^6)(3s^2\ 3p^6)$

σ_{Cl} = 6 x 0.35 + 8 x 0.85 + 2 x 1 = 10.9

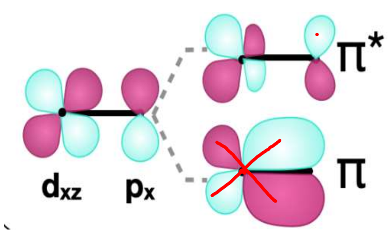
σ_{Cl^-}= 7 x 0.35 + 8 x 0.85 + 2 x 1 =11.25

- $Z^*{}_{Cl}$= 17-10.9 = 6.1
- $Z^*{}_{Cl^-}$= 17-11.25 = 5.75
- Since Z* of Cl is > than Z* of Cl^-, the size of Cl < Cl^-.

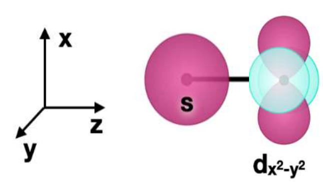
- **3. a.** Arrange these complexes $[V(CO)_6]^-$, $Cr(CO)_6$, or $[Mn(CO)_6]^+$ in order of increasing C-O bond length. Explain the trend. 1+2
- C-O bond length of $[Mn(CO)_6]^+ < Cr(CO)_6 < [V(CO)_6]^-$
- Negative charge on V increases its electron density and its capacity to back donate to the pi* orbital of CO, making its bond longer.
- +1 charge on Mn decreases its electron density, hence back-donation from Mn is relatively lower compared to the other two complexes. Hence, here CO will have the shortest bond length.

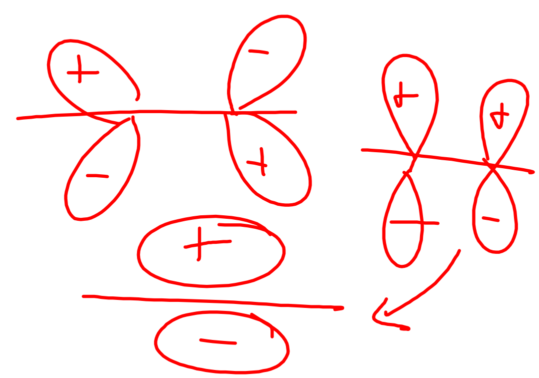
- **b.** Draw the orbitals and predict if the following pair of atomic orbitals will overlap to form molecular orbitals: 2

  - **i.** d_{xz} and p_x
  - ii. s and $d_{x^2-y^2}$
  - dxz will overlap with px orbital to form pi molecular orbital.



  - s and $d_{x^2-y^2}$ orbitals will not overlap.

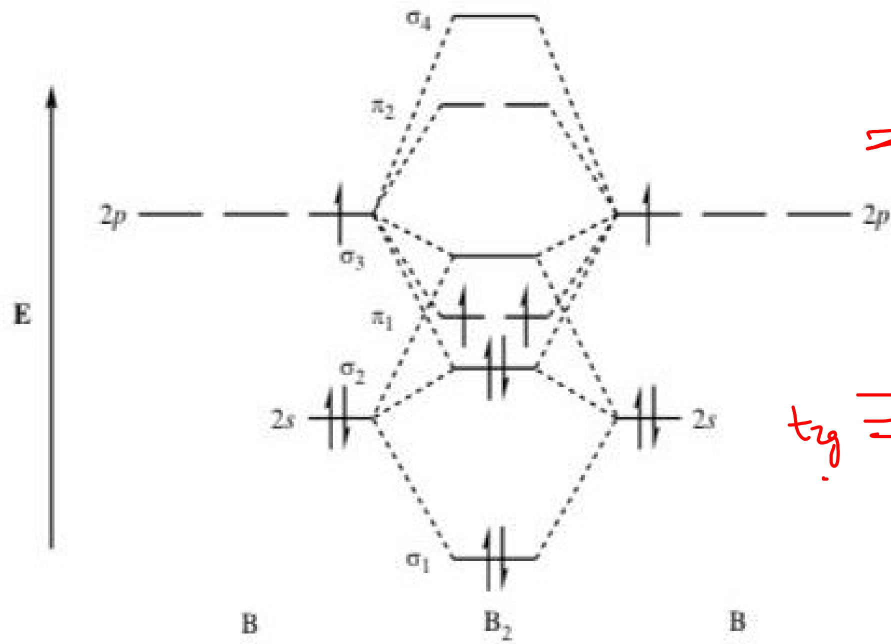


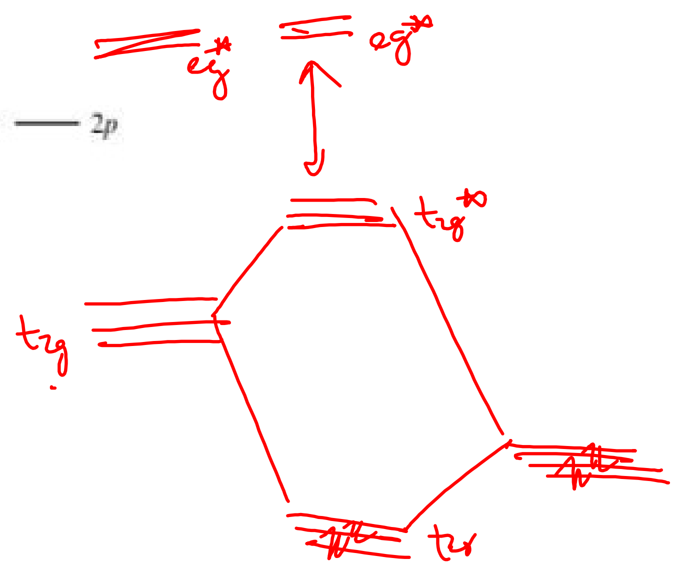


  - Here z is the internuclear axis.

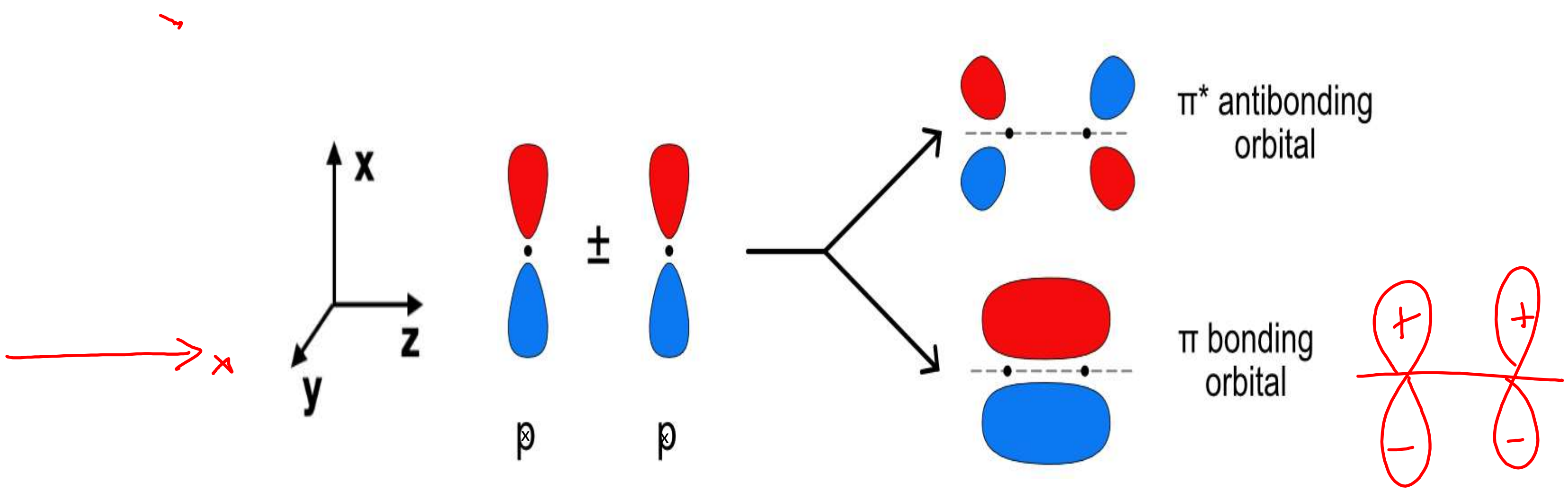
- **4.** **a.** Draw the molecular orbital energy level diagram of B_2 and comment on its magnetic behavior. 1+1

- Since B_2 has 2 unpaired electrons in its degenerate pi orbitals π_{2p_x} and π_{2p_y} orbitals B_2 is a paramagnetic molecule.





- **b.** Draw pictures of π_{2p_x} and $\pi^*_{2p_x}$ molecular orbitals. 3
- **i.** Draw and label the bond axis
- **ii.** Label g and u for these molecular orbitals.



- **5a.** The molecule $(CH_3)_2N\text{-}PF_2$ has two basic atoms, P and N. One is bound to B in a complex with BH_3, the other to B in a complex with BF_3. Decide which is which and state the reason for your decision.

  1+2

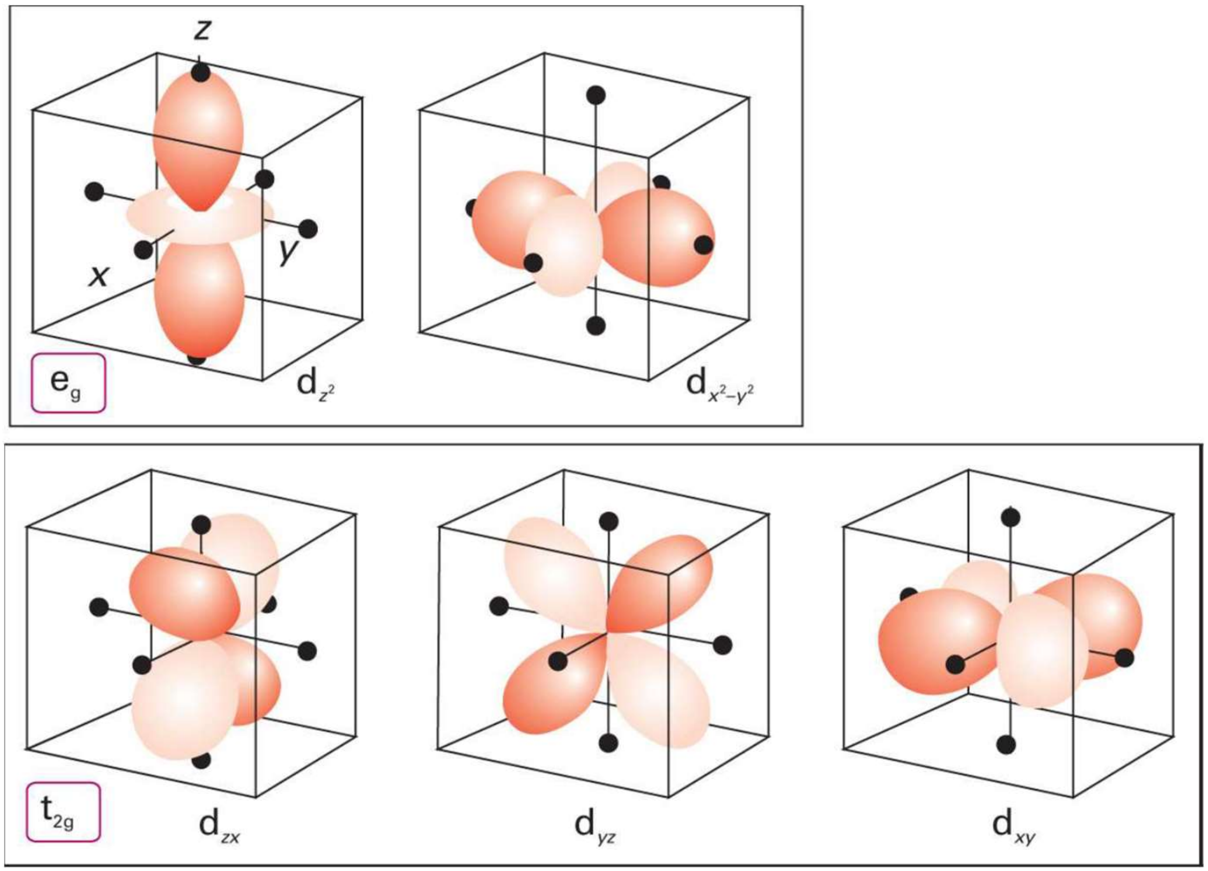
- P is a softer base than N. BH_3 is a softer acid than BF_3. Since hard acid prefers to bind to hard base, P will bind to B in BH_3 and N will bind to B in BF_3.

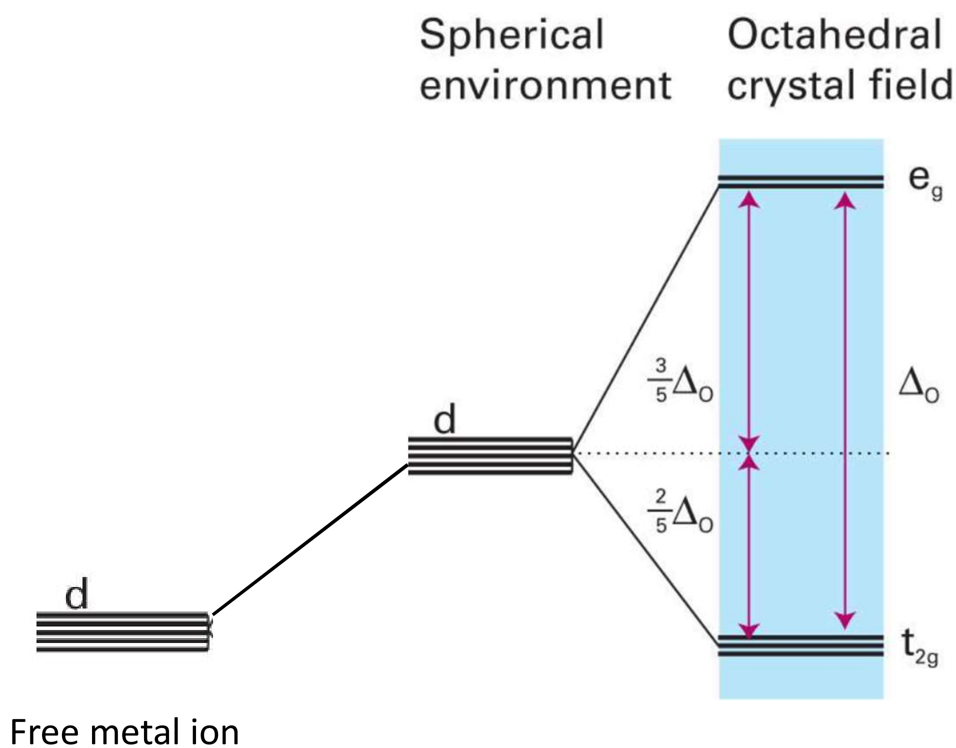
b. The lithium halides have solubility in water in the order LiBr > LiCl > LiF. Explain the trend. 2

- The strong hard–hard interaction in LiF overcomes the tendency of LiF to be solvated by water. The weaker hard–soft interactions between Li^+ and the other halides are not strong enough to prevent solvation, and these halides are more soluble than LiF.
- **Alternate explanation:** The high lattice energy of LiF surpass the hydration energy of Li^+ and F^- hence, it is insoluble. The other Li-halides, as the halides are larger, they have lower lattice energy and are more soluble.

Crystal Field Theory (CFT)

- Based on electrostatic model
- Ligand lone-pair is modelled as point negative charge
- Splitting of d-orbitals
- Correlate the optical spectra, thermodynamic stability and magnetic properties


z
x
y
e_g
d_{z^2}
$d_{x^2-y^2}$
t_{2g}
d_{zx}
d_{yz}
d_{xy}



Spherical environment
Octahedral crystal field
e_g
$\frac{3}{5}\Delta_O$
Δ_O
d
$\frac{2}{5}\Delta_O$
d
t_{2g}
Free metal ion

Δ_o depends on

i. the identity of the ligands

ii. Identity of the central metal ion and its oxidation state

Spectrochemical Series

$I^- < Br^- < Cl^- < OH^- < RCO_2^- < H_2O < NH_3 < en < CO, CN^-$

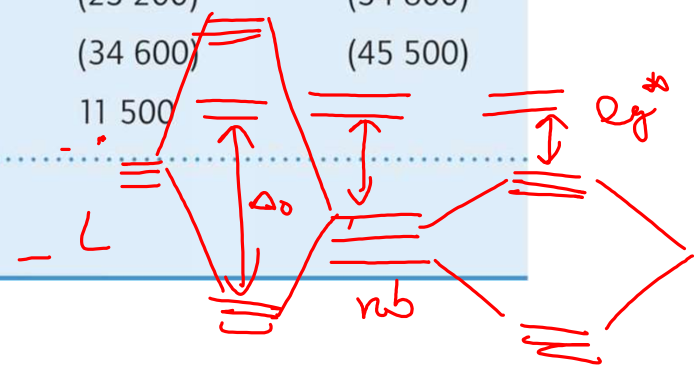
High spin	Low spin
Weak field	Strong field
Small Δ	Large Δ

Table 20.1 Ligand-field splitting parameters Δ_O of ML_6 complexes*

	Ions	Ligands Cl^-	H_2O	NH_3	en	CN^-
d^3	Cr^{3+}	13 700	17 400	21 500	21 900	26 600
d^5	Mn^{2+}	7500	8500		10 100	30 000
	Fe^{3+}	11 000	14 300			(35 000)
d^6	Fe^{2+}		10 400			(32 800)
	Co^{3+}		(20 700)	(22 900)	(23 200)	(34 800)
	Rh^{3+}	(20 400)	(27 000)	(34 000)	(34 600)	(45 500)
d^8	Ni^{2+}	7500	8500	10 800	11 500	

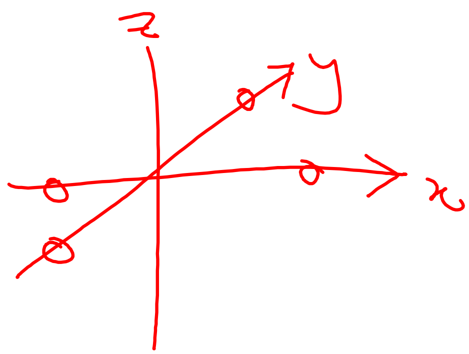
* Values are in cm^{-1}; entries in parentheses are for low-spin complexes.

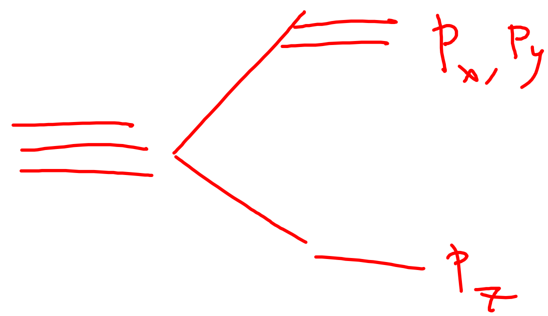
Source: H.B. Gray, *Electrons and chemical bonding.* Benjamin, Menlo Park (1965).



Crystal Field Stabilization Energy (CFSE)

Will there be any splitting
of the p orbitals?





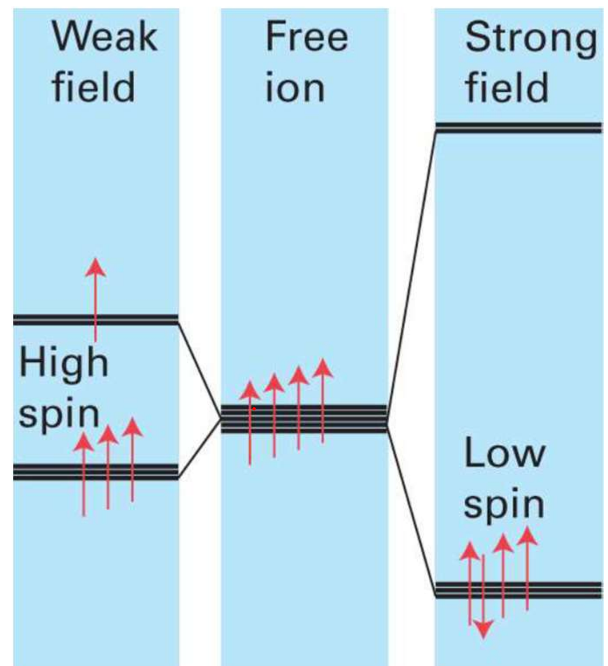


Table 1. Crystal Field Stabilization Energies (CFSE) for Octahedral Complexes

d^n	Spin State	Configuration	CFSE
d^1	–	$t_{2g}{}^{1}$	$-0.4\Delta_o$
d^2	–	$t_{2g}{}^{1+1}$	$-0.8\Delta_o$
d^3	–	$t_{2g}{}^{1+1+1}$	$-1.2\Delta_o$
d^4	high	$t_{2g}{}^{1+1+1}\ e_g{}^{1}$	$-0.6\Delta_o$
	low	$t_{2g}{}^{2+1+1}$	$-1.6\Delta_o + P$
d^5	high	$t_{2g}{}^{1+1+1}\ e_g{}^{1+1}$	0
	low	$t_{2g}{}^{2+2+1}$	$-2.0\Delta_o + 2P$
d^6	high	$t_{2g}{}^{2+1+1}\ e_g{}^{1+1}$	$-0.4\Delta_o$
	low	$t_{2g}{}^{2+2+2}$	$-2.4\Delta_o + 2P$
d^7	high	$t_{2g}{}^{2+2+1}\ e_g{}^{1+1}$	$-0.8\Delta_o$
	low	$t_{2g}{}^{2+2+2}\ e_g{}^{1}$	$-1.8\Delta_o + P$
d^8	–	$t_{2g}{}^{2+2+2}\ e_g{}^{1+1}$	$-1.2\Delta_o$
d^9	–	$t_{2g}{}^{2+2+2}\ e_g{}^{2+1}$	$-0.6\Delta_o$
d^{10}	–	$t_{2g}{}^{2+2+2}\ e_g{}^{2+2}$	0

e_g

t_{2g}

$-0.4+$

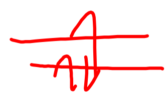
$-2.4\Delta_o + 2P$

- Pairing energies need to be taken into account only for pairing that is additional to the pairing that occurs in a spherical field.

Practice Problem:

1. Determine the LFSE for the following octahedral ions:

a. d^3



b. high-spin d^5

c. high-spin d^6

d. low-spin d^6

e. d^9

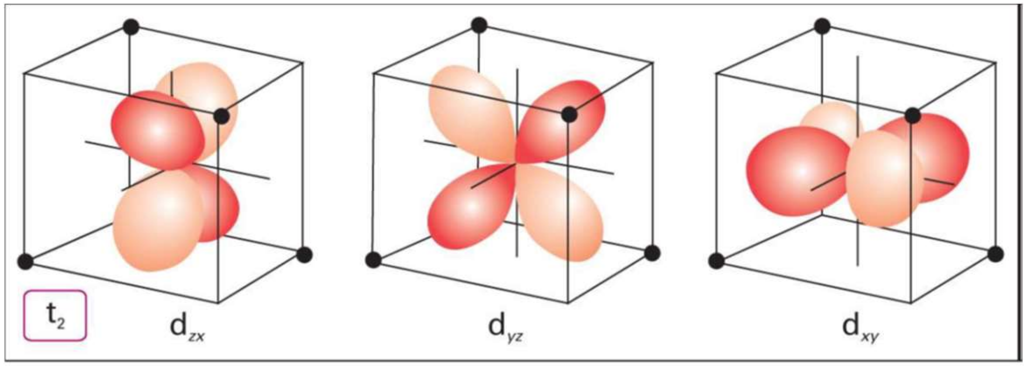
$-24\Delta_o + 18\Delta_o$

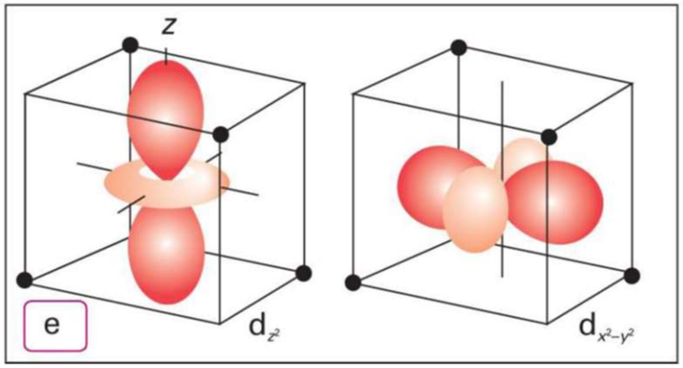
$= -0\ 6\ \Delta_o$

For octahedral complexes

- The strength of ligand-metal interaction is greater for metals having higher charges.
- Δ_o for 3 + ions is larger than for 2+ ions
- Metals from the second and third transition series form low-spin complexes more readily than metals from the first transition series.
- greater overlap between the larger 4d and 5d orbitals and the ligand orbitals
- decreased pairing energy due to the larger volume available for electrons in the 4d and 5d orbitals as compared with 3d orbitals.

Tetrahedral Geometry





d- orbital Splitting in Tetrahedral Geometry

- e orbitals lie below t_2 orbitals
- Only high-spin complex
- $\Delta_T \approx {}^4/_9\, \Delta_O$